Test for a Single Mean: One Sample t-test

Week 5 Lecture 3

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Hypothesis Testing

- **Aim:** Evaluate the strength of evidence against the null hypothesis
- **Steps:**
- Write null & alternative hypotheses
- Test the null hypothesis (2 options)
  - Simulation based approach
  - Theory based approach
- Get the null distribution
- Find p-value and/or standardized statistic
- Evaluate your evidence against the null/in favor of alternative hypothesis

Confidence Interval

- **Aim:** Provide an estimation to an unknown parameter.
- **Steps:**
- Find the sample statistic
- Compute margin of error
- Provide your interval estimation

	One Proportion – (Categorical variable)	One Mean – (Quantitative variable)
Graphical summary	Bar graph	Histogram, dotplot
Summary measures	$\hat{p}$ (sample proportion) n (sample size)	$\bar{x}$ (sample mean) s (sample standard deviation) n (sample size)
Statistic	$\hat{p}$ (sample proportion)	$\bar{x}$ (sample mean)
Parameter	π (population proportion or long-run proportion)	μ (population mean or long-run mean)
Null hypothesis	Ho: π = hypothesized proportion	Ho: μ = hypothesized mean
Alternative hypothesis	Ha: π > hypothesized proportion OR π < hypothesized proportion OR $\pi \neq$ hypothesized proportion	Ha: μ > hypothesized mean OR μ < hypothesized mean OR $\mu \neq$ hypothesized mean
Null distribution	Distribution of sample proportions	Distribution of sample means
Validity Conditions for Theory-based approach	Random sampling At least 10 successes and at least 10 failures in sample	Random Sampling Symmetric **population distribution** OR n > 30 AND **sample distribution** not strongly skewed
Test Statistic (or Standardized Statistic)	$z = \frac{\hat{p} - hypothesized\ MEANnull}{SDnull}$ OR $z = \frac{\hat{p} - hypothesized\ MEANnull}{\sqrt{\frac{\pi(1-\pi)}{n}}}$	$t = \frac{\bar{x} - hypothesized\ MEANnull}{SDnull}$ OR $t = \frac{\bar{x} - hypothesized\ MEANnull}{\frac{s}{\sqrt{n}}}$
95% Confidence Interval (2SD Method)	$\hat{p} \pm 2\ SDnull$ or $\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$\bar{x} \pm 2\ SDnull$ or $\bar{x} \pm 2\frac{s}{\sqrt{n}}$
SD_{null} (or SE)	SD_{null} from applet OR $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	SD_{null} from applet OR $SE(\bar{x}) = \frac{s}{\sqrt{n}}$

Backpack Weights

Backpack Weights

- Carrying a heavy backpack can be a source of chronic, low-level trauma and can cause long-lasting shoulder, neck, and back pain.
- The American Academy of Orthopedic Surgeons recommends that a backpack not weigh more than 10% to 15% of the wearer's body weight.
- College students at a public university in California (about 20,000 enrollment) wondered if college students at their university were following these backpack weight guidelines.

Backpack Weights

- **STEP 1: Ask a research question**
- Do university students carry backpacks that weigh on average something different than the recommended 10% of their body weight?
- Let's take a closer look at this study

Backpack Weights

- **STEP 2: Design a study and collect data**
- Each of 4 consecutive days, student researchers set up a scale at a different location on their campus. At each location, data was gathered on students who agreed to participate until they had data on 25 students at that location (100 total students).
- Each student had their backpack weight recorded to the nearest pound.
- Each student filled out a short survey that asked their weight in pounds, what college they were in, their major, their year in school, whether they were a graduate or undergraduate student, their total course units, and whether they experienced any back problems.

Backpack Weights

- What are the observational units?
- *The observational units are the 100 students.*
- What measured variables are categorical?
- *College, major, year in school, back problems.*
- What measured variables are quantitative?
- *The backpack weight (lbs), the student weight (lbs), the number of course units.*

Backpack Weights

- Other variables calculated
- Percentage of student's body weight the backpack is calculated as a ratio
- Spread sheet/data sheet

backpackw	bodyweigh		college	Major	Year	UG/G	Units	Back.probl	Ratio
9	125		CSM	Bio	3	U	13	Yes	0.072
8	195		CLA	Philosophy	5	U	12	No	0.041026
10	120		CLA	GRC	4	U	14	Yes	0.083333
6	155		ENG	CSC	6	G	0	No	0.03871
8	180		ENG	EE	2	U	14	No	0.044444
5	240		CLA	History	0	G	0	No	0.020833
8	170		CAED	CM	3	U	15	No	0.047059
4	185		CAED	ARCE	5	U	18	Yes	0.021622
5	130		CSM	Bio	4	U	14	No	0.038462
2	120		CSM	Bio	5	U	8	No	0.016667
8	135		OCB	Bus	3	U	15	No	0.059259
21	160		ENG	ME	5	U	12	No	0.13125
11	170		ENG	CPE	4	U	16	No	0.064706

Hypotheses

- State the null and alternative hypotheses
  - **Null hypothesis:** The unknown long run mean ratio of backpack weight to body weight is 0.10
  - **Alternative hypothesis:** The unknown long run mean ratio of backpack weight to body weight is different from 0.10

Parameters

- We can also use a symbol for the unknown parameter we are trying to estimate.
- The parameter of interest is:
  - μ = the unknown long-run mean ratio of backpack weight to body weight

Hypotheses

- Mu (μ) is the parameter for population mean ratio.
- Using this symbol, we can rewrite the hypotheses as follows:

$$\mathbf{H_0}: \mu = 0.10$$

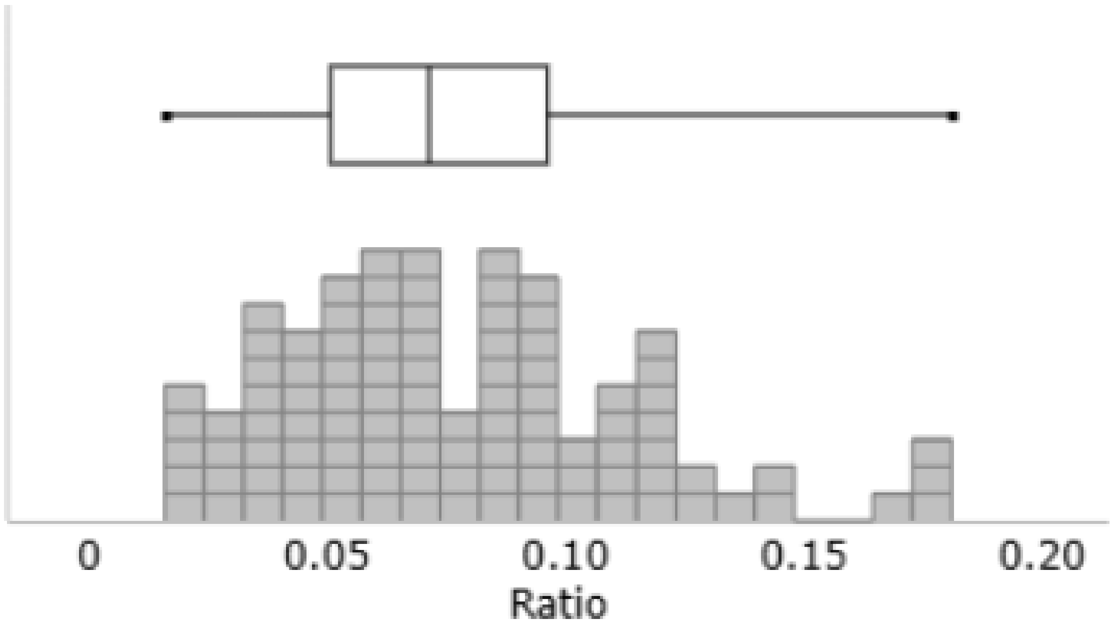
$$\mathbf{H_a}: \mu \neq 0.10$$

Hypotheses

Remember

- The hypotheses are about the parameter value in the population, not just for these 100 students.
- Hypotheses are always about populations or processes, not the sample data.

Step 3: Explore the data



	Sample size	Mean	SD	Min.	First Quart.	Med.	Third Quart.	Max.
Ratio	100	0.078	0.037	0.016	0.052	0.071	0.096	0.181

Results

- The sample mean was different than 0.10
- What are two possible explanations for why we observed a sample mean that was different from 0.10?
- #1 The population mean ratio really is 0.10 and our sample mean of 0.078 from 100 students happened by random chance.
- #2 The population mean ratio is different from 0.10 and that is why our sample mean ratio was different from 0.10.

Step 4: Draw inferences beyond the data

- Is it *possible* to get a mean ratio of 0.078 even if in the population the ratio is 0.10?
- Yes, it’s possible, how likely though?
- Test statistic and p-value will tell how likely it is to get a mean ratio that different from 0.10.

Test Statistic

- We will use a t-test to conduct a hypothesis test about a population mean. The structure, reasoning, and interpretation are the same as with all hypothesis tests.
- Can we carry out a t-test on these data?
- YES! Our sample size is greater than 30 and the sample data is roughly symmetric (only slightly skewed to the right)

$$t = \frac{(\bar{x} - \mu)}{s/\sqrt{n}}$$

$$t = \frac{(0.078 - 0.10)}{0.037/\sqrt{100}} = -5.946$$

How to Find p-value?

- Let's use the applet to find the p-value.
- Very small p-value. Reject Ho.

Confidence Interval

- We also see that an approximate 95% confidence interval for the mean ratio is:

$$0.078 \pm 2(0.004)$$ or about 0.071 to 0.085

- This means we are 95% confident that the mean ratio of backpack weight to body weight is between 0.071 to 0.085
- This means that university students at this university in California carry backpacks that on average are between 7.1% to 8.5% of their total body weight
- What does it mean for the entire interval to be below 0.10 (below 10%)?

Step 5: Formulate Conclusions

Was the sample randomly selected from a larger population?

- No, the sample was not randomly sampled from a larger population of university students. However, if you believe that university students in California are not different in their body composition or the amount of weight they carry in their backpacks compared to other university/college students across the US, then we can generalize to all college/university students in the US.

Were the observational units randomly assigned to treatments?

- No, there was no random assignment in this study, so we cannot make any causal conclusions.

Conclusions

- While we have strong evidence that the mean ratio is different from 0.10 (10%), is the difference impressive?
- Recall that for children in elementary school and secondary school, pediatric surgeons recommended that the ratio of backpack weight to body weight not exceed 10% to 15%.
- The average ratio of 0.08 (7.8%) is well below this recommendation.
- It does seem that college/university students are carrying more reasonable weights in their backpacks.

Step 6: Look back and ahead

- **Looking back:** Did anything about the design and conclusions of this study concern you? In particular, are there things that could have been done to give a better chance finding strong evidence of a true difference between the two groups?
  - Always nice to have a larger sample size. A random sample. Researchers should try to replicate their results. These replicated studies could be at different colleges/universities across the U.S.
- **Looking ahead:** What should the researchers' next steps be to fix the limitations or build on this knowledge?
  - See if these results hold when other variables are looked at. For example do these ratios hold within subpopulations like males and females, or between different majors. Could backpack weights be decreasing due to e-books (all books on one laptop).